

P-Set IV

POLI 784 - Spring 2026

2026-02-28

Complete all of the questions below together with your group members. Please submit both (1) a compiled PDF or HTML file with your responses and (2) the **R Markdown**, **Quarto**, etc. file containing both your responses and code as a group. You are encouraged to use **R Studio** projects and virtual environments to complete this p-set, although it is not required. Similarly, you are encouraged to use consistent style and linters when writing code.

Question 1

Consider the following simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + e_i$$

- a. (*1 point*) What do β_0 , β_1 , and e_i represent in the regression model and how do we interpret their values, respectively? What does β_1 *not* being indexed by i and β_1 being *multiplied* with x_i tell us about the structural restrictions of OLS regression estimates?
- b. (*0.5 points*) What would be the values of β'_0 and β'_1 (and why) if the explanatory variable values in the above regression were transformed using ...
 - i. ... $x' = x - 10$ and y was regressed on x' .
 - ii. ... $x' = 10x$ and y was regressed on x' .
 - iii. ... $x' = 10(x - 1) = 10x - 10$ and y was regressed on x' .
- c. (*0.5 points*) What would be the values of β''_0 and β''_1 (and why) if the response variable values in the above regression were transformed using ...
 - i. ... $y'' = y + 10$ and y'' was regressed on x .
 - ii. ... $y'' = 5y$ and y'' was regressed on x .
 - iii. ... $y'' = 5(y + 2) = 5y + 10$ and y'' was regressed on x .

- d. (0.5 points) In general, how are the β_0 and β_1 coefficients of OLS regression models affected when x_i or y_i are shifted or scaled by some constant c , as in the above examples? Do any of these changes affect the *substantive* conclusions about the relationship between x_i and y_i we would draw from the regression analysis? Why or why not?

Question 2

- a. (1 point) Consider the bivariate structural model and the associated assumption stated below.

$$Y_i = \mu + \tau D_i + \varepsilon_i$$

$$E[\varepsilon_i | D_i] = 0$$

- i. What do $\hat{\varepsilon}_i$ (sometimes represented as e_i) and ε_i represent in the regression context, respectively?
 - ii. When does $E[\hat{\varepsilon}_i | D_i] = 0$ hold in OLS regression?
 - iii. When does $E[\varepsilon_i | D_i] = 0$ hold in the structural model and what does it imply for our OLS estimates of τ ?
- b. (0.5 points) Suppose the above bivariate regression model is misspecified by virtue of omitting a relevant variable U_i . Derive the expression of $Y_i - \bar{Y}$ under the correct model specified below. (Note: We are deriving this expression to aid us in the next part of the problem. $E[\varepsilon_i | D_i, U_i] = 0$ is impertinent to this current step and can be ignored for now.)

$$Y_i = \mu + \tau D_i + \beta U_i + \varepsilon_i$$

$$E[\varepsilon_i | D_i, U_i] = 0$$

- c. (0.5 points) The expression representing the *bivariate* OLS estimator of τ is provided to you below. Substitute the expression you derived for $Y_i - \bar{Y}$ in the *multivariate* case in 2b into the bivariate OLS estimator and simplify the expression. (Note: In the final simplified expression, $\hat{\tau}$ should equal the sum of three terms in this form: $\hat{\tau} = \tau + \beta A + B$.)

$$\hat{\tau} = \frac{\sum_{i=1}^N (Y_i - \bar{Y})(D_i - \bar{D})}{\sum_{i=1}^N (D_i - \bar{D})^2}$$

- d. (0.5 points) Since the third term, B , converges to zero as $N \rightarrow \infty$, the second term, βA – which itself converges to $\beta \frac{Cov[D_i, U_i]}{Var[D_i]}$ – is the mathematical expression of omitted variable bias in bivariate regression (when there is only one omitted confounder, U_i). Looking at the expression you derived for βA (or the one it converges to), what has to be mathematically true for omitted variable bias to be 0?

Question 3

In this exercise, you will be working with data from Santoro and Broockman (2022). In their experiment, the researchers invited subjects to participate in an online video chat with another respondent, who was an out-partisan. During their conversation, subjects talked about what their perfect day would be like. Subjects were then randomly assigned into either the treatment group ($D_i = 1$), where they were informed that the other subject was an out-partisan, or the control group ($D_i = 0$), where they received no additional information. The outcome variable is post-treatment warmth toward out-partisans. You can find the data required for this exercise in `data.csv`.

- a. (0.5 points) Compute both an ATE point estimate using the Neyman estimator and the corresponding analytical Neyman variance estimate. Then estimate the ATE and the HC2 variance estimate using the OLS regression estimator without any covariates. Compare the two – what do you notice and how do you explain this?
- b. (1 point) Fit a new OLS regression model, this time controlling for all covariates in the data. Test the Frisch-Waugh-Lovell (FWL) theorem by implementing the algorithm to compute the ATE estimate and showing that the coefficient for $\hat{\tau}$ equals that of D_i in the multiple OLS regression by printing them to console.
- c. (1 point) Use the residuals attained from the FWL algorithm to plot a scatterplot and regression line of the relationship between Y_i and D_i after controlling for all the covariates in the model. Plot a second scatterplot and regression line using the original Y_i and D_i to show the bivariate descriptive relationship between the two without controlling for third factors. What difference do you notice, if any?
- d. (1 point) Fit a Lin regression model including all covariates. Plot the coefficients for the treatment and the associated analytical 95% CIs based on the HC2 variances of the bivariate OLS, multiple OLS, and Lin regression models in a single coefficient plot.
 - i. Compare their ATE point estimates and their performance in terms of efficiency.
 - ii. Under which circumstances could we consider the point estimate $\hat{\tau}$ from any one of the three models an unbiased causal estimate of the ATE?
 - iii. What is your assessment of the plausibility of these circumstances holding true in this case? Why are they likely or unlikely to be true when applying these estimators to these data?
- e. (0.5 points) Test the joint null hypothesis that the effect of D_i on Y_i equals zero for both men and women by conducting a Wald test based on the Lin regression model. Interpret the results.
- f. (1 point) Estimate the moderator effect of *pre-treatment* warmth toward out-partisans by (1) plotting the treatment effect estimate from the Lin regression model across the range of the moderator variable and (2) fitting a kernel regression using `interflex()` to estimate the

moderator effect of pre-treatment warmth on the treatment effect and plotting the results. Interpret the results.